

Yarone Meir Tokayer

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EDUCATION

Yale University

Ph.D., Physics, Advisor: Frank van den Bosch

Thesis: “Probing the dynamical structure of dark matter halos using N-body and analytical techniques”

M.S. and M.Phil

exp. 2027

May 2023

Columbia University

M.A., Philosophical Foundations of Physics, Advisor: David Z. Albert

Thesis: “Probability in Everettian Quantum Mechanics”

New York, NY

Feb. 2020

The Cooper Union

B.S., Engineering, Minor in Mathematics, *summa cum laude*

Senior Project: “Muscle Denervation and Neurotensin as a Therapeutic Target for ALS”

New York, NY

May 2014

SKILLS AND LANGUAGES

- **Programming languages:** Python, MATLAB, C++, HTML/CSS, FORTRAN
- **Software & tools:** NASA HEASoft, Mathematica, Latex, LabVIEW, Excel
- **N-body code:** `gyrfalcON`, Agama
- **Python Libraries:**
Astronomy: Astropy, Stingray, PyXSPEC
Data and Visualization: Pandas, NumPy, Matplotlib
- **Telescope data:** *Chandra*, *Swift*, *NuSTAR*, *NICER*
- **Ground telescope observing experience:**
Palomar: Double Spectrograph, TripleSpec
Keck: NIRC2+AO, OSIRIS+AO
- **Design tools:** Arduino, Microchip PIC, AutoCAD, SolidWorks, laser cutting
- **Spoken languages:** English (native), Hebrew (fluent), Yiddish (basic), German (basic)

RESEARCH EXPERIENCE

Modeling the dynamical structure of dark matter halos

Graduate Research Assistant, PI: Frank van den Bosch

Theoretical and computational astrophysics.

- Developed gravothermal fluid codes to model stellar mass segregation and SIDM core collapse.
- Implemented idealized N-body code to model CDM halo growth and compare with cosmological predictions.

Quantifying Selection Effects in Deep-Field AGN Surveys Using BASS

Graduate Research Assistant, PIs: Michael Koss, C. Megan Urry

Extra-galactic high energy astrophysics.

- Developed simulation data pipeline to generate and fit high-*z* Chandra AGN spectra.
- Spectral analysis of 2800 simulated spectra to quantify Chandra bias.
- Work published in *ApJ*; poster presented at the Jan. 2023 AAS meeting.

Constraining Dark Matter Profiles with GGSL Measurements

Graduate Research Assistant, PI: Priyamvada Natarajan

Data-driven cosmology.

Jun. 2024–Present

Yale University

Jul. 2021–Aug. 2024

Yale University

Jan. 2022–May 2024

Yale University

- Developed fitting algorithms for cluster DM subhalo profiles from lensing measurements.
- Derived empirical c-M relation, furthering Λ CDM tension.
- Work published in *ApJ*.

Galactic X-ray astronomy data analysis

Mar. 2020–Dec. 2020

Research Assistant, PIs: Charles Hailey, Kaya Mori

Columbia University

Galactic high energy astrophysics.

- Timing analysis of NICER observations of AR Scorpii. Sub- μ s precision spin measurement; found pulsed non-thermal emission in X-ray band.
- Imaging analysis of NuSTAR observations of the “Eel Nebula.” Discovered synchrotron burnoff effect in the PWN and evidence of a shock feature. Work published in *ApJ*.
- Spectral and timing analysis of NuSTAR observations of HESS J0632+057. Work published in *ApJ*.

Assembly and testing of detector modules for GAPS experiment

Aug. 2019–Feb. 2020

Research Assistant, PI: Charles Hailey

Columbia University

Experimental particle astrophysics.

- Fabrication, testing, and assembly of Si(Li) detector array modules for GAPS flight (2022); aims to detect antimatter evidence of DM annihilation in the galactic halo.

ALS pathology in mouse and cell culture models

Sep. 2013–May 2014

Research Assistant, PI: Christopher Henderson; Supervisor: Dima Yudin

Columbia Univ. Med. Ctr

Molecular and Cellular Neurobiology.

- Senior capstone project: ALS pathology in mouse and cell culture models. Immunohistochemistry to measure denervation over time. Optimization of embryonic stem cell-derived motor neuron cultures to test trophic factors.

Lung Perfusion Bioreactor

Jan. 2011–May 2011

Undergraduate Researcher, PIs: Eric Lima, Gordana Vunjak-Novakovic

Cooper Union/Columbia University

- Designed and built prototype used for testing on swine lung.

Med. Ctr

PUBLICATIONS AND POSTERS

A complete list of publications can also be found on [Google Scholar](#).

Refereed Journal Publications

1. Tokayer, Y. M., Koss, M., Urry, C. M., et al., *ApJ*, 982, 134 (2025).
doi:[10.3847/1538-4357/adb8c9](#)
2. Tokayer, Y. M., Dutra, I., Natarajan, P., et al., *ApJ*, 970, 143 (2024).
doi:[10.3847/1538-4357/ad51fd](#)
3. Guolo, M., Gezari, S., Yao, Y., et al. (incl. Tokayer, Y. M.), *ApJ*, 966, 160, (2024).
doi:[10.3847/1538-4357/ad2f9f](#)
4. Burgess, D., Mori, K., Gelfand, J. D., et al. (incl. Tokayer, Y. M.), *ApJ*, 930, 148, (2022).
doi:[10.3847/1538-4357/ac650a](#)
5. Tokayer, Y. M., An, H., Halpern, J. P., et al., *ApJ*, 923, 17, (2021).
doi:[10.3847/1538-4357/ac2c6a](#)

Conference Posters

6. Tokayer, Y. M., Koss, M., Urry, C. M., et al., Quantifying Selection Effects in Deep-Field AGN Surveys with BASS. In: *241st Meeting of the AAS*, January 8–12, 2023. Seattle, WA.
[Link to poster.](#)

7. Woo, J., An, H., Burgess, D., et al. (incl. **Tokayer, Y. M.**), Multi-wavelength Study of PeVatron Candidate Pulsar Wind Nebulae. In: *AAS/High Energy Astrophysics Division 19th Annual Meeting*, March 13–17, 2022. Pittsburgh, PA.
8. Saffold, N., **Tokayer, Y. M.**, Mori, K., A NICER X-ray View of White Dwarf Pulsar AR Scorpii. In: *237th Meeting of the AAS*, January 10–15, 2021. Virtual.
[Link to poster](#).

ACADEMIC PRESENTATIONS

Invited Talks

9. “Core-collapsed SIDM halos as massive SMBH seeds,” Hebrew University Cosmolunch; Apr. 24, 2025. Jerusalem, Israel.
10. “The Unified AGN Model in X-ray observations,” Columbia University High-Energy Astrophysics Meeting; Jan. 28, 2022. Virtual.

Conference Talks

11. “Core-collapsed SIDM halos as massive SMBH seeds,” [YCIU Workshop](#); May 30, 2025; New Haven, CT.
12. “Using BASS to Detect Obscuration Bias and Test AGN Fitting Models for Low Count Data,” BASS2024: New Horizons for Understanding Nearby AGN; April 22, 2024; Virtual.
13. “Quantifying AGN Selection Effects in the Chandra COSMOS-Legacy Survey with BASS,” [Accretion History of AGN \(AHA\) III Workshop](#); December 15, 2023; Miami, FL.
14. “Quantifying Selection Effects in Deep-Field AGN Surveys,” [New England Regional Quasar and AGN Meeting \(NERQUAM\)](#); May 6, 2022; Storrs, CT.
15. Saffold, N., **Tokayer, Y. M.**, “A NICER X-ray View of White Dwarf Pulsar AR Scorpii,” [Spring 2021 NICER Data Analysis and Science Workshop](#); May 13, 2021. Virtual.
[Video of presentation](#).

ACCEPTED PROPOSALS

16. Co-Investigator. Ref #GN-2024A-Q-138 for Gemini GMOS Observations. *Resolving Mrk 248: A Potential Triple AGN with a Hidden 500 pc Dual AGN*.
PI: Michael Koss
17. Co-Investigator. Proposal 8087 for NuSTAR Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at $z=0.2-0.4$* .
PI: Michael Koss
18. Co-Investigator. Proposals 90296 and 094349 for XMM-Newton Observations. *A Survey of the Most Luminous Hard X-ray Selected Obscured Quasars at $z=0.2-0.4$* .
PI: Michael Koss

CONFERENCES, WORKSHOPS, AND SUMMER SCHOOLS ATTENDED

- **AI Winter School**
Brown University. January 6–9, 2026. Virtual.
- **Yale Center for the Invisible Universe (YCIU) Workshop**
Yale University. May 30, 2025. New Haven, CT.

- **Accretion History of AGN (AHA) III Workshop**
University of Miami. December 14–17, 2023. Miami, FL.
- **Cosmology Summer School**
University of Michigan. June 5–9, 2023. Ann Arbor, MI.
- **241st Meeting of the AAS**
January 8–12, 2023. Seattle, WA.
- **New England Regional Quasar and AGN Meeting (NERQUAM)**
University of Connecticut. May 6, 2022. Storrs, CT.
- **NICER Data Analysis and Science Workshop**
May 13, 2021. Virtual.

TEACHING

Leitner Observatory and Planetarium

Planetarium Presenter January 2024–Present

- Write and present planetarium shows using night sky simulation for groups spanning pre-K to adult
- Design and deliver classroom presentations about astronomy and outer space for school groups K-12
- Run telescope demonstrations for evening groups

Yale University

Course design

- Designed, wrote lecture slides and problem sets for undergraduate course in Summer/Fall 2024
Cosmology (ASTR 170); Instructor: Priyamvada Natarajan

Graduate Teaching Fellow

- Graduate Cosmology (PHYS 6000) Spring 2026
- Introduction to Mathematical Methods of Physics (PHYS 301, PHYS 4000) Fall 2022, Fall 2025
- Graduate Statistical Physics I (PHYS 512) Spring 2024, Spring 2025
- Graduate Classical Mechanics (PHYS 500) Fall 2024
- University Physics for the Life Sciences (PHYS 171) Spring 2023
- General Physics Laboratory II (PHYS 166L/S166) Spring 2022, Summer 2022
- General Physics Laboratory I (PHYS 165L) Fall 2021

SAR High School

Chemistry Teacher Spring 2021

- Taught four 10th grade chemistry sections spanning three levels from remedial to honors

Physics Teacher and Advisor Fall 2014–Spring 2019

- Designed curriculum and taught 4 physics courses on two tracks for 11th and 12th grade
- Taught 11 grade Jewish Philosophy course, 9 and 11 grade Jewish text study
- 10th grade advisor
- Designed and oversaw engineering-related elective and co-curricular programming
- Coached robotics team that won second place in [robotics competition](#) at the Technion in Haifa, Israel

Naaleh High School for Girls

STEM Teacher Fall 2019–Spring 2020

- Wrote curriculum and taught a course in computer programming and engineering design
- Designed and instructed Python coding co-curricular; club was a member of [Girls Who Code](#)

The Cooper Union

Teaching Assistant Fall 2011

- Introductory Physics Lab (PH291)

COMMUNITY ENGAGEMENT AND OUTREACH

Service

Graduate Student Mentor , Yale Undergraduate Mentorship Initiative	2025–Present
Co-Founder and Organizer , History and Foundations of Physics Reading Group	2022–Present
Board of Trustees , Slifka Center for Jewish Life at Yale	2023–2024
Graduate Student Representative , Physics Department Faculty Hiring Committee	Spring 2023

Outreach

Astronomy on Tap <i>-Astronomy nights at a New Haven bar, featuring trivia and talks by members of the Yale Astronomy Department.</i>	Fall 2022
Super Science Showdown , Yale Open Labs <i>-Interactive science events for students in grades 6-8 in New Haven County.</i>	Spring 2022
Engineers as Teachers , Iridescent and Cooper Union <i>-Wrote and built interactive lesson plans on the topic of sound and music -Lessons were taught at family science nights in local NYC middle school</i>	Spring 2011

Public Talks

1. “Are we alone? The new search for other worlds and habitable planets beyond the Solar System,” SAR High School; Apr. 17, 2024; Riverdale, NY.
[Video of talk.](#)
2. “The hype is real: what we are already learning from the most powerful telescope in human history,” SAR High School; Jan. 4, 2023; Riverdale, NY.
[Video of talk.](#)

AWARDS AND SCHOLARSHIPS

- **3-Minute Thesis Competition Finalist**
Yale University Graduate School of Arts and Sciences, 2026
- **Leigh Page Award for Excellence in Graduate Student Teaching**
Yale University Physics Department, 2025
- **Teacher Award**
2017 Robotraffic Competition at the Technion in Haifa, Israel, as coach to the SAR High School team
- **Entrance Scholarship**
Philosophical Foundations of Physics Program, Columbia University, Fall 2016
- **Tau Beta Pi**
Engineering Honors Society
- **Goodman Prize**
Cooper Union, Spring 2013
- **Dean’s List**
Cooper Union, all semesters